Project Title: RouteGuard Predict Supply Chain Problems Before They Happen

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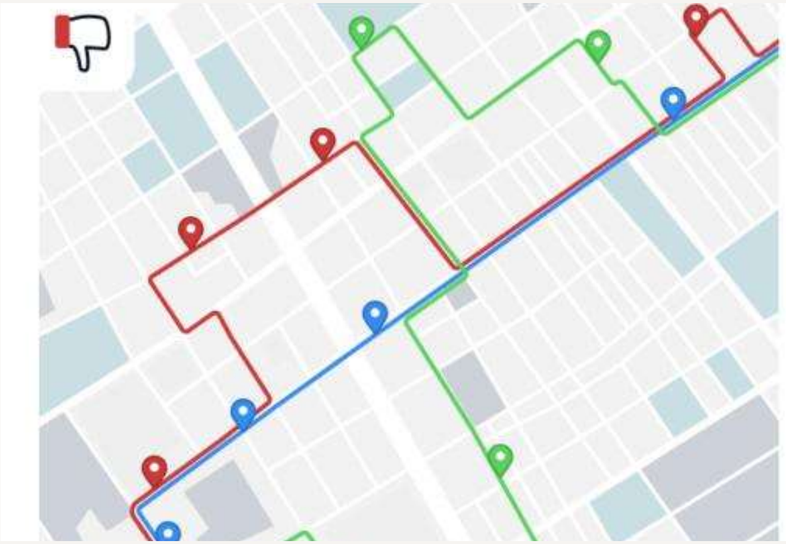
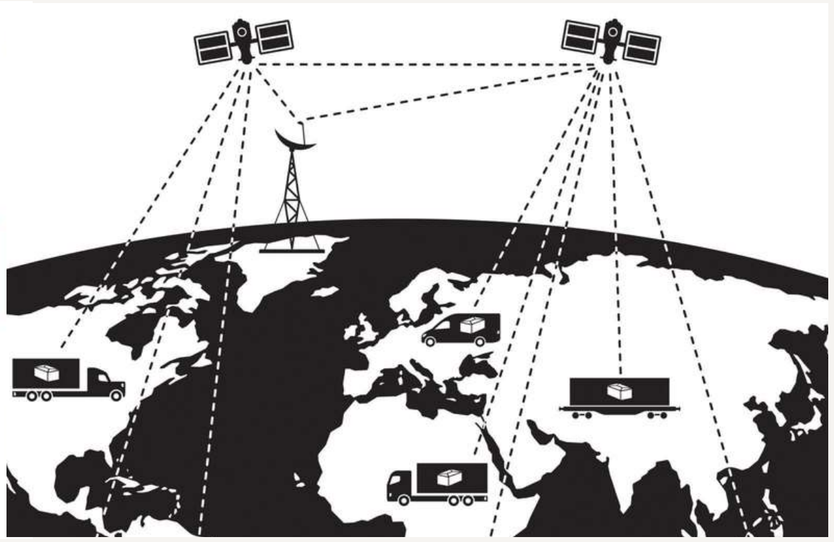
Problem Statement: Why Supply Chains Keep Failing

- Millions of shipments move globally every day, but companies lack real-time visibility
- Problems like storms, traffic, and port congestion are detected only after delays occur
- Current systems are reactive, not proactive, so actions are taken too late
- Data exists (weather, traffic, ports) but is not integrated for decision-making
- One manager cannot monitor hundreds of shipments across multiple regions
- Existing tools only show location, not future risks or disruptions
- This leads to huge losses, delays, poor coordination, and inefficiency
- This leads to delays, losses, poor coordination, and inefficient supply chain operations

Solution Overview

- RouteGuard continuously monitors shipments using real-time data like weather, traffic, and port conditions
- It predicts potential disruptions before they occur instead of reacting after delays
- System calculates a risk score (0–100) for each shipment based on multiple factors
- When risk is high, it automatically suggests better alternate routes with comparison
- Managers can view insights, understand causes, and take action with one click
- All stakeholders (shipper, driver, receiver) stay connected with real-time updates
- System learns from past shipments to improve accuracy over time





Technical Implementation

- System collects real-time data from weather, traffic, ship tracking, and port conditions
- Raw data is processed into meaningful features like weather severity, traffic congestion, and port risk
- ML models (XGBoost, Random Forest, LSTM) predict risk score, delay time, and future risk trends
- System evaluates alternate routes and suggests best option based on time, cost, and risk
- Real-time alerts are sent instantly using live communication (Socket-based updates)
- Data is managed using PostgreSQL (structured) and MongoDB (real-time), with Redis for fast access
- Dashboard provides live tracking, risk visualization, alerts, and one-click decision support

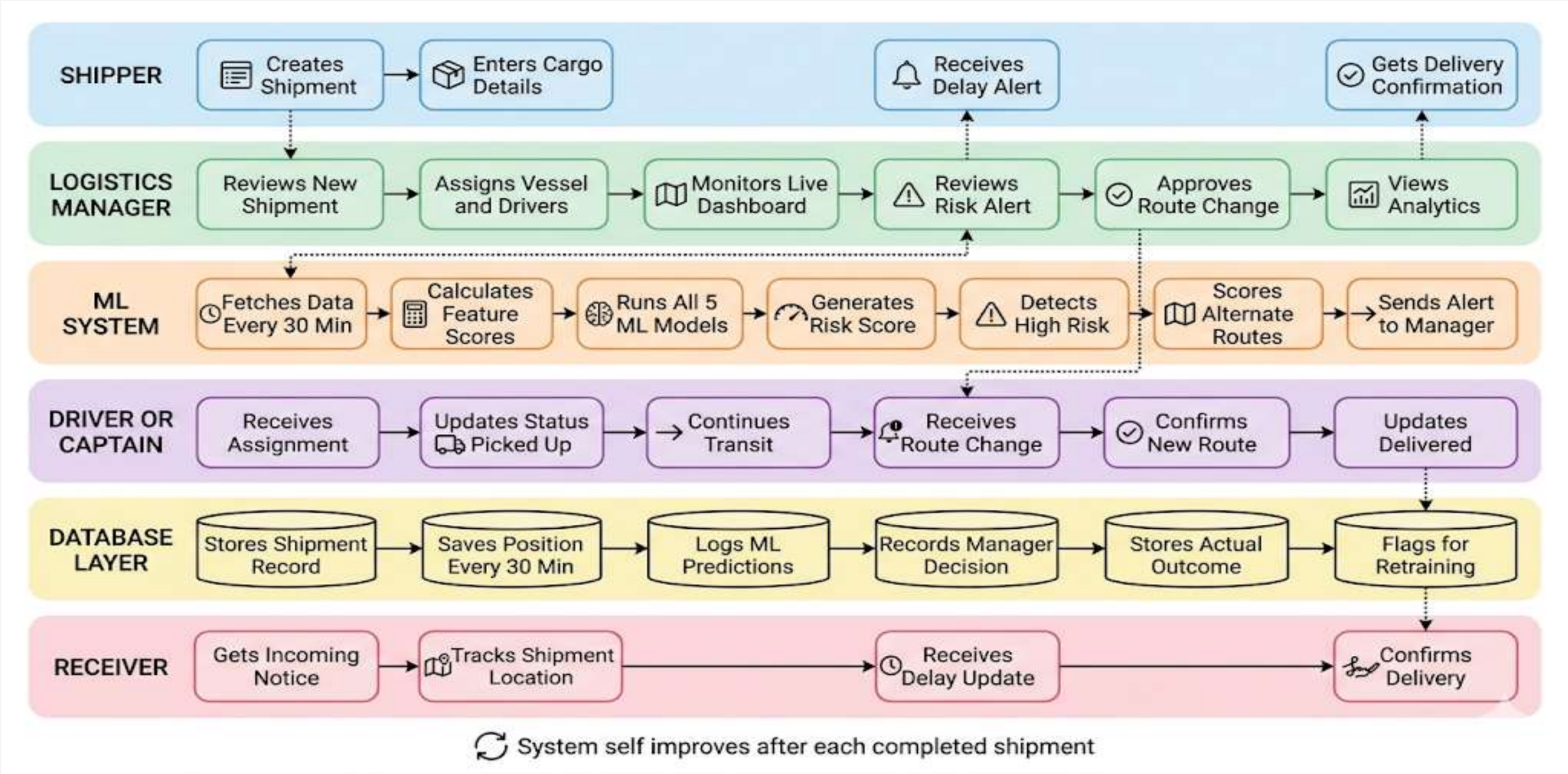
Market Feasibility

- Global supply chain market is huge ($26.5B+) and growing rapidly
- Logistics industry faces $1.5 trillion losses yearly due to disruptions
- 79% of disruptions can be predicted using better data and analytics
- Target customers include logistics companies, manufacturers, and insurers
- Strong revenue model with SaaS subscriptions and data API services
- High willingness to pay due to large financial impact of delays
- Huge market opportunity with scalable global adoption

Impact:

- Reduces shipment delays and prevents major financial losses
- Saves cost for companies by avoiding disruptions early
- Improves delivery reliability and customer satisfaction
- Reduces food waste and ensures timely delivery of medicines
- Helps small businesses avoid unexpected losses
- Reduces carbon emissions by optimizing routes and avoiding air freight
- Moves industry from reactive to proactive decision-making

Work Flow:


SHIPPER
Creates Shipment
Enters Cargo Details
Receives Delay Alert
Gets Delivery Confirmation
LOGISTICS MANAGER
Reviews New Shipment
Assigns Vessel and Drivers
Monitors Live Dashboard
Reviews Risk Alert
Approves Route Change
Views Analytics
ML SYSTEM
Fetches Data Every 30 Min
Calculates Feature Scores
Runs All 5 ML Models
Generates Risk Score
Detects High Risk
Scores Alternate Routes
Sends Alert to Manager
DRIVER OR CAPTAIN
Receives Assignment
Updates Status Picked Up
Continues Transit
Receives Route Change
Confirms New Route
Updates Delivered
DATABASE LAYER
Stores Shipment Record
Saves Position Every 30 Min
Logs ML Predictions
Records Manager Decision
Stores Actual Outcome
Flags for Retraining
RECEIVER
Gets Incoming Notice
Tracks Shipment Location
Receives Delay Update
Confirms Delivery
System self improves after each completed shipment

Future Scope & Next Steps

- Improve system using real-world datasets and increase prediction accuracy over time
- Build mobile application for drivers with real-time updates and notifications
- Integrate IoT sensors to monitor cargo conditions like temperature, humidity, and damage
- Add AI-based document checking to prevent customs delays before shipment
- Introduce carbon tracking to measure and optimize environmental impact
- Develop supplier risk monitoring to detect issues before shipment starts
- Move towards automation with smart rerouting and minimal human intervention